Module-1

Solution of Algebraic and Transdental equations

Method-1

1) Bisection method / Interval halfing / Bolzano's method

- Let g. $f(x) = 0$ be the given equation, find two points A and B. $f(a)$ and $f(b)$ are opposite sign. ~~but f(a) be~~.
- het $f(a)$ be positive and $f(b)$ be negative
- Then $x_0 = \frac{a+b}{2}$
- Then find the sign of $f(x_0)$.
- If $f(x_0)$ is -ve, then $x_1 = \frac{a+x_0}{2}$
- If $f(x_0)$ is +ve, then $x_1 = \frac{x_0+b}{2}$
- Find the signe of $f(x_1)$
- Continue the procedure till 2 consecutive values are same

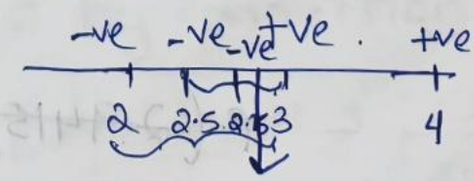
root of equation

Q) Assuming that $x^3 - 9x + 1 = 0$ lies in the interval (2,4). Find the root of the equation using bisection method

→ het $f(x) = x^3 - 9x + 1$

$f(2) = 2^3 - 9 \times 2 + 1 = -9$

$f(4) = 29$



Let $a = 2$, $b = 4$

$x_0 = \frac{2+4}{2} = 3$

$f(x_0) = f(3) = 3^3 - 9 \times 3 + 1 = 1$ +ve.

$x_1 = \frac{2+3}{2} = 2.5$

$f(2.5) = -5.875 = -ve$

$x_2 = 2.75$

$f(2.75) = -2.9$

$x_3 = \frac{2.75+3}{2} = 2.875$

$f(2.875) = -1.111 = -ve$

$x_4 = \frac{2.875+3}{2} = 2.9375$

$f(2.9375) = -0.09 = -ve$.

$x_5 = \frac{2.9375+3}{2} = 2.9688$.

$f(2.9688) = 0.44 = +ve$

$x_6 = \frac{2.9375+2.9688}{2} = 2.9532$

$f(2.9532) = 0.177 = +ve$

$x_7 = 2.94584$

$f(2.9454) = 0.04 = +ve$.

$x_8 = \frac{2.9375+2.9454}{2} = 2.9415$

~~$f(2.9415) = 2.9375+2.9415$~~

$f(2.9415) = -0.022 = -ve$.

$x_9 = \frac{2.9454+2.9415}{2} = 2.9435$

$f(2.9435) = 0.011 = +ve$.

$x_{10} = \frac{2.9435+2.9415}{2} = 2.9425$.

$f(2.9425) = -5.43 \times 10^{-3} = -ve$.

$x_{11} = \frac{2.9435+2.9425}{2} = 2.9430$.

$f(2.9430) = 3 \times 10^{-3} = +ve$.

$x_{12} = \frac{2.9430+2.9425}{2} = 2.9428$.

$f(2.9428) = -3.4 \times 10^{-4} = -ve$

$x_{13} = \frac{2.9430+2.9428}{2} = 2.9429$

$f(2.9429) = 1.35 \times 10^{-3} = +ve$.

$x_{14} = \frac{2.9429+2.9428}{2} = 2.9429$

* Find +ve root of $x - \cos x = 0$ by Bisection method

→ $f(x) = x - \cos x = 0$

$f(0) = 0 - \cos 0 = 0 - 1 = -1$

$f(1) = 1 - \cos(1) = 0.4597$.

The root lies in $(0,1)$.

let $a = 0$ $b = 1$.

$x_0 = \frac{a+b}{2} = \frac{0+1}{2} = 0.5$

$f(x_0) = 0.5 - \cos(0.5)$

$= -0.3776$ -ve.

$x_1 = \frac{1+0.5}{2} = 0.75$

$f(0.75) = 0.75 - \cos(0.75) = 0.0183 = +ve$.

$x_2 = \frac{0.75+0.5}{2} = 0.625$

$f(0.625) = -0.1850 = -ve$.

$x_3 = \frac{0.625+0.75}{2} = 0.6875$

$f(0.6875) = 0.6875 - \cos(0.6875) = -0.0853 = -ve$.

$x_4 = \frac{0.6875+0.75}{2} = 0.7188$

$f(0.7188) = 0.7188 - \cos(0.7188) = -0.0338$ -ve

$x_5 = \frac{0.7188+0.75}{2} = 0.7344$

$f(0.7344) = 0.7344 - \cos(0.7344) = -7.83\times10^{-3} = -ve$.

$x_6 = \frac{0.7344+0.75}{2} = 0.7422$

$f(0.7422) = 0.7422 - \cos(0.7422)$

$= 5.21\times10^{-3} = +ve$

$x_7 = \frac{0.7344+0.7422}{2} = 0.7383$

$f(x_7) = f(0.7383) = 0.7383 - \cos(0.7383)$

$= -1.3137\times10^{-3}$ (-ve)

$x_8 = \frac{0.7383+0.7422}{2} = 0.7403$

$f(x_8) = 0.7403 - \cos(0.7403)$

$f(0.7403) = 2.03\times10^{-3}$ (+ve).

$x_9 = \frac{0.7383+0.7403}{2} = 0.7393$

$f(0.7393) = 0.7393 - \cos(0.7393) = 3.6\times10^{-4}$ (+ve)

$x_{10} = \frac{0.7383+0.7393}{2} = 0.7388$.

$f(0.7388) = 0.7388 - \cos(0.7388) = -4.77\times10^{-4}$.

$= (-ve)$

$x_{11} = \frac{0.7388+0.7393}{2} = 0.7390$

$f(0.7390) = 0.7390 - \cos(0.7390) = -1.42\times10^{-4} = -ve$

$x_{12} = \frac{0.7390+0.7393}{2} = 0.7392$

$f(0.7392) = 0.7392 - \cos(0.7392) = 1.92\times10^{-4} = +ve$

$x_{13} = \frac{0.7390+0.7392}{2} = 0.7391$

$f(0.7391) = 0.7391 - \cos(0.7391) = 2.4\times10^{-5} = +ve$

$$x_{14} = \frac{0.7390+0.7391}{2} = 0.7391$$

+ve root $= 0.7391$

Iteration method/Method of successive approximation

→ Suppose we want the approximate root of the equation. $f(x)=0$ ① write the given equation in the form. $x=\phi(x)$ ② Assuming x_0 to be the starting approximate value to the actual root, setting $x=x_0$ in RHS of equation ②, we get $\boxed{x_1=\phi(x_0)}$ again setting $x=x_1$ on RHS of ② $\boxed{x_2=\phi(x_1)}$ continue the procedure till 2 consecutive values are equals

Note

Let (a,b) be the interval Containing the root of the given equation. Then the sufficient conditions for convergence is $\boxed{|\phi'(x)|<1}$ for all x in (a,b)

① Find a real root of the equation $e^{x-bx}=0$ $e^x - 3x = 0$ by method of iteration.

→ $f(x) = e^x - 3x$

$f(0) = e^0 - 3\times0 = 1$ (+ve)

$f(1) = e^1 - 3 = -0.28$ (-ve)

interval $(0,1)$

Root lies between 0 and 1

Let $x = \frac{e^x}{3} = \phi(x)$

Let $x_0 = 0.8$.

$x_1 = \phi(x_0) = \frac{e^{0.8}}{3} = 0.7418$

$x_2 = \phi(x_1) = \frac{e^{0.7418}}{3} = 0.6999.$

$x_3 = \phi(x_2) = \frac{e^{0.6999}}{3} = 0.6712$

$x_4 = \phi(x_3) = \frac{e^{0.6712}}{3} = 0.6522.$

$x_5 = \phi(x_4) = \frac{e^{0.6522}}{3} = 0.6399$

$x_6 = \phi(x_5) = \frac{e^{0.6399}}{3} = 0.6321$

$x_7 = \phi(x_6) = \frac{e^{0.6321}}{3} = 0.6272$

$x_8 = \phi(x_7) = \frac{e^{0.6272}}{3} = 0.6241$

$x_9 = \phi(x_8) = \frac{e^{0.6241}}{3} = 0.6222$

$x_{10} = \phi(x_9) = \frac{e^{0.6222}}{3} = 0.6210$

$x_{11} = \phi(x_{10}) = \frac{e^{0.6210}}{3} = 0.6203$

$x_{12} = \phi(x_{11}) = \frac{e^{0.6203}}{3} = 0.6198$

$x_{13} = \phi(x_{12}) = \frac{e^{0.6198}}{3} = 0.6195$

$x_{14} = \phi(x_{13}) = \frac{e^{0.6195}}{3} = 0.6193$

$x_{15} = \phi(x_{14}) = \frac{e^{0.6193}}{3} = 0.6192$

$x_{16} = \phi(x_{15}) = \frac{e^{0.6192}}{3} = 0.6191$

$x_{17} = \phi(x_{16}) = \frac{e^{0.6191}}{3} = 0.6191$

Solution = 0.6191

Q) Solve $\cos x = 3x - 1$ by iteration method.

$f(x) = \cos x - 3x + 1$

$f(0) = 2$. (+ve)

$f(1) = -1.45$ (-ve)

$\phi(x) = \frac{\cos x + 1}{3}$

$\phi'(x) = \frac{-\sin x + 0}{3} = \left|\frac{-\sin x}{3}\right| < 1$

$x_0 = 0.6$.

$x_1 = \phi(x_0) = \frac{\cos(0.6) + 1}{3} = 0.6084$

$x_2 = \phi(x_1) = \frac{\cos(0.6084) + 1}{3} = 0.6069$.

$x_3 = \phi(x_2) = \frac{\cos(0.6069) + 1}{3} = 0.6071$.

$x_4 = \phi(x_3) = \frac{\cos(0.6071) + 1}{3} = 0.6071$

Solution is 0.6071

Newton Raphson Method (Method of tangent)

Let x_0 meet an approximate root of $f(x) = 0$. The equation of tangent at $(x_0, f(x_0))$ is $y - f(x_0) = f'(x_0)(x - x_0)$. This tangent cuts the x axis at $x = x_1$ to get the point put $y = 0$ in the equation of tangent $-f(x_0) = f'(x_0)(x - x_0)$.

or $x_1 = x = x_0 - \frac{f(x_0)}{f'(x_0)}$. Take the point $(x, f(x))$ on the curve. Take find the tangent at that point

It cuts the x axis at $x = x_2$ where $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$

Continue the procedure till two consecutive points are equal

Q) Find the root between 0 and 1 of $x^3 = 6x - 4$ correct to 5 decimal places by Newton Raphson method

→ Let $f(x) = x^3 - 6x + 4$

$f(0) = 4$ +ve

$f(1) = -1$ (-ve)

$f'(x) = 3x^2 - 6$

Let $x_0 = 0.5$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$f(0.5) = 1.125$

$f'(0.5) = -5.25$

$$x_1 = 0.5 - \frac{1.125}{-5.25} = 0.71429$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = \frac{2x_n^3 - 4}{3x_n^2 - 6}$$

$$= \frac{2(0.71429)^3 - 4}{3(0.71429)^2 - 6} = 0.731890$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$= x_n - \frac{x_n^3 - 6x_n + 4}{3x_n^2 - 6}$$

$$= \frac{3x_n^3 - 6x_n - x_n^3 + 6x_n - 4}{3x_n^2 - 6}$$

$$= \frac{2x_n^3 - 4}{3x_n^2 - 6}$$

$$= \frac{2x_0 - 4}{3x_0^2 - 6}$$

$$x_3 = \frac{2(0.73190^3) - 4}{3(0.73190)^2 - 6} = 0.73205$$

$$x_4 = \frac{2(0.73205^3) - 4}{3(0.73205^2) - 6} = 0.73205$$

Solution = 0.73205

Q) Find the positive root of $x = \cos x$ using Newton's method

→ $f(x) = x - \cos x$

$f(0) = 0 - \cos 0 = -1$

$f(1) = 1 - \cos(1) = 0.4597$

$f'(x) = 1 + \sin x$

$x_0 = 0.7$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$= x_n - \frac{(x_n - \cos x_n)}{1 + \sin x_n}$$

$$= \frac{x_n + x_n \sin x_n - x_n + \cos x_n}{1 + \sin x_n}$$

$$x_{n+1} = \frac{x_n \sin x_n + \cos x_n}{1 + \sin x_n}$$

$$x_1 = \frac{0.7\sin 0.7 + \cos 0.7}{1+\sin(0.7)} = 0.73943$$

$$x_2 = \frac{0.7394 \sin 0.7394 + \cos 0.7394}{1+\sin(0.7394)}$$

$$= 0.7391$$

$$x_3 = \frac{0.7391 \sin 0.7391 + \cos 0.7391}{1+\sin(0.7391)}$$

$$= 0.7391$$

Solution is 0.7391

Q) Find an iterative formula to find $\sqrt{N}$ where N is a positive number and hence find ~~roots~~ $\sqrt{5}$ using Newton Raphson method

→ let $x = \sqrt{N}$

~~let x~~ $f(x) = x - \sqrt{N}$

$f'(x) = 1$

$x_{n+1} = x_n - \frac{x_n - \sqrt{N}}{1}$ } not possible.

$x^2 = N$

$x^2 - N = 0$

let $f(x) = x^2 - N$

$f'(x) = 2x$.

$x_{n+1} = x_n - \frac{x_n^2 - N}{2x_n}$

$= \frac{2x_n^2 - x_n^2 + N}{2x_n}$.

$x_{n+1} = \frac{x_n^2 + N}{2x_n}$

$x_0 = 2.3$.

$x_1 = \frac{2.3^2 + 5}{2 \times 2.3} = 2.2370$

$x_2 = \frac{2.2370^2 + 5}{2 \times 2.2370} = 2.2361$

$x_3 = \frac{2.2361^2 + 5}{2 \times 2.2361} = 2.2361$

Solution is 2.2361

* Using Newton's Raphson method, Find iterative formula for

→ a) reciprocal of N

b) $\sqrt[3]{N}$

c) $\sqrt[p]{N}$

→ (a) $x = \frac{1}{N}$

$x - \frac{1}{N} = 0$

$N = \frac{1}{x}$.

$f'(x) = \frac{1}{x^2}$

$\frac{1}{x} - N = 0$

$x_n - \frac{f(x_n)}{f'(x_n)}$

$f(x) = \frac{1}{x} - N$ —— ①

$x_n - x_n + \frac{1}{N} \Rightarrow \frac{1}{N}$

$f'(x) = -\frac{1}{x^2}$

$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

$= x_n - \frac{\left[\frac{1}{x_n} - N\right]}{\frac{-1}{x_n^2}}$ =

$= x_n \times \frac{-1}{x_n^2} - \frac{1}{x_n} + N$

$\frac{-1}{x_n^2}$

$= \frac{-\frac{2}{x_n} + N}{\frac{-1}{x_n^2}} = \left(\frac{2}{x_n} - N\right) \times x_n^2$

$x_{n+1} = 2x_n - Nx_n^2 = x_n(2 - Nx_n)$

$\boxed{x_{n+1} = x_n(2 - Nx_n)}$

b) $3\sqrt{N}$

$x = 3\sqrt{N}$

$x^3 = N$

$x^3 - N = 0$.

$f(x) = x^3 - N$.

$f'(x) = 3x^2$.

Newton Raphson formula.

$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$.

$= x_n - \frac{(x_n^3 - N)}{3x_n^2} = \frac{3x_n^3 - x_n^3 + N}{3x_n^2}$

$= \frac{2x_n^3 + N}{3x_n^2} = \frac{1}{3}\left[2x_n + \frac{N}{x_n^2}\right]$

c) $x = P\sqrt{N}$

$x^P = N$

$x^P - N = 0$.

$f(x) = x^P - N$

$f'(x) = Px^{P-1}$.

$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{(x_n^P - N)}{Px_n^{P-1}}$

$= \frac{Px_n x_n^{P-1} - x_n^P + N}{Px_n^{P-1}}$

$$= \frac{P x_n^P - x_n^P + N}{P x_n^{P-1}} \qquad P-P+1$$

$$= \frac{(P-1)x_n^P + N}{P x_n^{P-1}} = \frac{(P-1)x_n^P}{P x_n^{P-1}} + \frac{N}{P x_n^{P-1}}$$

$$\boxed{x_{n+1} = \frac{1}{P}\left[(P-1)x_n + \frac{N}{x_n^{P-1}}\right]}$$

Secant Method or Chord method

$$x_{k+1} = \frac{x_{k-1} f_k - x_k f_{k-1}}{f_k - f_{k-1}}, \quad k = 1, 2, 3, \ldots$$

Q) A root of the equation $xe^x - 1 = 0$ lies in the interval $(0.5, 1)$. Determine this root correct to three decimal places using secant method.

→ Let $f(x) = xe^x - 1$

$x_0 = 0.5$

$x_1 = 1$

$k = 1$.

$f(x_0) = f_0 = 0.5e^{0.5} - 1 = -0.1756$

$f(x_1) = f_1 = 1e^1 - 1 = 1.7183$

~~$x_2 = \frac{x_0 f_1 - x_1 f_0}{}$~~ $x_{k+1} = \frac{x_{k-1} f_k - x_k f_{k-1}}{f_k - f_{k-1}}$

$k = 1$.

$$x_2 = \frac{x_0 f_1 - x_1 f_0}{f_1 - f_0}$$

~~$= \frac{0.5 \times -0.1756 - 1 \times}{}$~~

$$= \frac{0.5 \times 1.7183 - 1 \times -0.1756}{1.7183 - -0.1756}$$

$$= 0.546$$

$$f(x_2) = f_2 = 0.546(e)^{0.546} - 1$$

$$= -0.057$$

$k = 2$.

$$x_3 = \frac{x_1 f_2 - x_2 f_1}{f_2 - f_1}$$

$$= \frac{1 \times -0.057 - 0.546 \times 1.7183}{-0.057 - 1.7183}$$

$$x_3 = 0.561$$

$$f_3 = 0.561(e)^{0.561} - 1$$

$$= -0.017$$

$k = 3$.

$$x_4 = \frac{x_2 f_3 - x_3 f_2}{f_3 - f_2}$$

$$= \frac{0.546 \times -0.017 - 0.561 \times -0.057}{-0.017 - -0.057}$$

$$= 0.567$$

$f_4 = 0.567e^{0.567} - 1$

$= -3.958 \times 10^{-4}$

$= -3.959 \times 10^{-4}$

$k = 4$

$$x_5 = \frac{x_3 f_4 - x_4 f_3}{f_4 - f_3}$$

$$= \frac{0.561 \times -3.959 \times 10^{-4} - 0.567 \times -0.017}{-3.959 \times 10^{-4} - -0.017}$$

$= 0.567$

solution is 0.567

H.W

Q) Use secant method to estimate the root of

$f(x) = e^{-x} - x$.

find $x_0, x_1 \Rightarrow f(x_0), f(x_1)$ (opposite sign)

$f_0 = f(0) = e^{0} - 0 = 1$

$f_1 = f(1) = e^{-1} - 1 = -0.632$ } opposite sign

$x_0 = 0 \quad x_1 = 1$

$$x_{k+1} = \frac{x_{k-1} f_k - x_k f_{k-1}}{f_k - f_{k-1}}$$

$k = 1$

$$x_2 = \frac{x_0 f_1 - x_1 f_0}{f_1 - f_0}$$

$$= \frac{1 \times -0.632 - 1 \times 1}{-0.632 - 1} = 0.6127$$

$f(x_2) = f_2 = e^{-0.6127} - 0.6127$

$= -0.0708$

$$x_3 = \frac{x_1 f_2 - x_2 f_1}{f_2 - f_1}$$

$$= \frac{1 \times -0.0708 - 0.6127 \times -0.632}{-0.0708 - -0.632}$$

$= 0.5638$

$f(x_3) = e^{-0.5638} - 0.5638 = 5.2426 \times 10^{-3}$

$$x_4 = \frac{x_2 f_3 - x_3 f_2}{f_3 - f_2}$$

$$= \frac{0.6127 \times 5.2426 \times 10^{-3} - 0.5638 \times -0.0708}{5.2426 \times 10^{-3} + 0.0708}$$

$x_4 = 0.5671$

$f(x_4) = e^{-0.5671} - 0.5671 = 6.18 \times 10^{-5}$

$$x_5 = \frac{x_3 f_4 - x_4 f_3}{f_4 - f_3}$$

$$= \frac{0.5628 \times 6.78\times10^{-5} - 0.5671 \times 5.2426\times10^{-3}}{6.78\times10^{-5} - 5.2426\times10^{-3}}$$

$$= 0.5671$$

Solution is 0.5671

imp

Gauss Seidel Method

→ $AX = B$.

$a_1x + b_1y + c_1z = d_1 \Rightarrow x = \frac{1}{a_1}[d_1 - b_1y - c_1z]$

$a_2x + b_2y + c_2z = d_2 \Rightarrow y = \frac{1}{b_2}[d_2 - a_2x - c_2z]$

$a_3x + b_3y + c_3z = d_3 \Rightarrow z = \frac{1}{c_3}[d_3 - a_3x - b_3y]$.

$|a_1| > |b_1| + |c_1|$ {diagonally dominant}

$|b_2| > |a_2| + |c_2|$

$|c_2| > |a_3| + |b_3|$

I^st iteration

let $y=0, z=0$

$x^{(i)} = \frac{1}{a_1}[d_1]$

$y^{(i)} = \frac{1}{b_2}[d_2 - a_2x^{(i)}]$

$z^{(i)} = \frac{1}{c^3}[d_3 - a_3x^{(i)} - b_3y^{(i)}]$

IInd iteration

$x^{(ii)} = \frac{1}{a_i}[d_1 - b_1y^{(i)} - c_1z^{(i)}]$

$y^{(ii)} = \frac{1}{b_2}[d_2 - a_2x^{(ii)} - c_2z^{(i)}]$

$z^{(ii)} = \frac{1}{c^3}[d_3 - a_3x^{(ii)} - b_3y^{(ii)}]$

Q) $x + y + 54z = 110$

$27x + 6y - z = 85$

$6x + 15y + 2z = 72$.

Using Gauss Seidel correct to three decimal places

→ Rearranging eqns to get diagonally dominant

$27x + 6y - z = 85$.

$6x + 15y + 2z = 72$.

$x + y + 54z = 110$.

$x = \frac{1}{27}[85 - 6y + z]$ —①

$y = \frac{1}{15}[72 - 6x - 2z]$ —②

$z = \frac{1}{54}[110 - x - y]$ —③.

I^st iteration

put $y=0, z=0$ in eq ①.

$x^{(i)} = \frac{1}{27} \times 85 = 3.148$

$$y^{(i)} = \frac{1}{15}[72 - 6x3.148] = 3.541$$

$$z^{(i)} = \frac{1}{54}[110 - 3.148 - 3.541] = 1.913$$

II nd iteration

$$x^{(ii)} = \frac{1}{27}[85 - 6y^{(i)} + z^{(i)}]$$

$$= \frac{1}{27}[85 - 6x3.541 + 1.913] = 2.432$$

$$y^{(ii)} = \frac{1}{15}[72 - 6x^{(ii)} - 2z^{(i)}]$$

$$= \frac{1}{15}[72 - 6x2.432 - 2x1.913] = 3.573$$

$$z^{(ii)} = \frac{1}{54}[110 - x^{(ii)} - y^{(ii)}]$$

$$= \frac{1}{54}[110 - 2.432 - 3.572] = 1.926$$

III rd iteration

$$x^{(iii)} = \frac{1}{27}[85 - 6x3.572 + 1.926]$$

$$= 2.426$$

$$y^{(iii)} = \frac{1}{15}[72 - 6x2.426 - 2x1.926]$$

$$= 3.573$$

$$z^{(iii)} = \frac{1}{54}[110 - 2.426 - 3.572]$$

$$= 1.926$$

IV th iteration

$$x^{(iv)} = \frac{1}{27}[85 - 6x3.573 + 1.926]$$

$$= 2.425$$

$$y^{(iv)} = \frac{1}{15}[72 - 6x2.425 - 2x1.926] = 3.573$$

$$z^{(iv)} = \frac{1}{54}[110 - 2.425 - 3.573] = 1.926$$

V th iteration

$$x^{(v)} = \frac{1}{27}[85 - 6x3.573 + 1.926] = 2.425$$

$$y^{(v)} = \frac{1}{15}[72 - 6x2.425 - 2x1.926] = 3.573$$

$$z^{(v)} = \frac{1}{54}[110 - 2.425 - 3.573] = 1.926$$

Solution $x = 2.425, y = 3.573, z = 1.926$

H.W Solve $28x + 4y - z = 32$

$x + 3y + 10z = 24$

$2x + 17y + 4z = 35$ using gauss seidel method

Interpolation

→ Rearranging equations

$$28x + 4y - z = 32$$
$$2x + 17y + 4z = 35$$
$$x + 3y + 10z = 24$$

I^st iteration

$y = 0 \quad z = 0$

$x = \frac{1}{28} \times 32 = 1.1429$

$y = \frac{1}{15}[35$

$$x = \frac{1}{28}[32 - 4y + z] \quad -①$$

$$y = \frac{1}{17}[35 - 2x - 4z] \quad -②$$

$$z = \frac{1}{10}[24 - x - 3y] \quad -③$$

I^st iteration

put $y = 0, z = 0$ in eq ①.

$$x^{(i)} = \frac{32}{28} = 1.1429.$$

put $x = 1.1429 \quad z = 0$ in eq ②

$$y^{(i)} = \frac{1}{17}[35 - 2 \times 1.1429] = 1.9244$$

put $x = 1.1429 \quad y = 1.9244$ in eq ③

$$z^{(i)} = \frac{1}{10}[24 - 1.1429 - 3 \times 1.9244] = 1.7084$$

IInd iteration

$$x^{(ii)} = \frac{1}{28}[32 - 4y^{(i)} + z^{(i)}] = \frac{1}{28}[32 - 4 \times 1.9244 + 1.7084]$$

$$= 0.92890$$

$$y^{(ii)} = \frac{1}{17}[35 - 2x^{(ii)} - 4z^{(i)}] = \frac{1}{17}[35 - 2 \times 0.9290 - 4 \times 1.7084]$$

$$= 1.5476$$

$$z^{(ii)} = \frac{1}{10}[24 - x^{(ii)} - 3y^{(ii)}] = \frac{1}{10}[24 - 0.9290 - 3 \times 1.5476]$$

$$= 1.8428$$

IIIrd iteration

$$x^{(iii)} = \frac{1}{28}[32 - 4y^{(ii)} + z^{(ii)}] = \frac{1}{28}[32 - 4 \times 1.5476 + 1.8428]$$

$$= 0.9876.$$

$$y^{(iii)} = \frac{1}{17}[35 - 2x^{(iii)} - 4z^{(ii)}] = \frac{1}{17}[35 - 2 \times 0.9876 - 4 \times 1.8428]$$

$$= 1.5090$$

$$z^{(iii)} = \frac{1}{10}[24 - x^{(iii)} - 3y^{(iii)}] = \frac{1}{10}[24 - 0.9876 - 3 \times 1.5090]$$

$$= 1.8485$$

→ ~~Rearranging equations~~

(IV)th iteration

$$x^{(IV)} = \frac{1}{28}\left[32 - 4y^{(III)} + z^{(III)}\right] = \frac{1}{28}[32 - 4\times 1.5090 + 1.8485]$$

$$= 0.9933$$

$$y^{(IV)} = \frac{1}{17}\left[35 - 2x^{iv} - 4z^{(III)}\right] = \frac{1}{17}[35 - 2\times 0.9933 - 4\times 1.8485]$$

$$= 1.5070$$

$$z^{(IV)} = \frac{1}{10}\left[24 - x^{iv} - 3y^{iv}\right] = \frac{1}{10}[24 - 0.9933 - 3\times 1.5070]$$

$$= 1.8486$$

(V)th iteration

$$x^{(V)} = \frac{1}{28}\left[32 - 4y^{(IV)} + z^{(IV)}\right] = \frac{1}{28}[32 - 4\times 1.5070 - 1.8486]$$

$$= 0.9936$$

$$y^{(V)} = \frac{1}{17}\left[35 - 2x^{(V)} - 4z^{iv}\right] = \frac{1}{17}[35 - 2\times 0.9936 - 4\times 1.8486]$$

$$= 1.5070$$

$$z^{(V)} = \frac{1}{10}\left[24 - x^{(V)} - 3y^{(V)}\right] = \frac{1}{10}[24 - 0.9936 - 3\times 1.5070]$$

$$= 1.8485$$

(VI)th iteration

$$x^{(VI)} = \frac{1}{28}\left[32 - 4y^{(V)} + z^{(V)}\right] = \frac{1}{28}[32 - 4\times 1.5070 + 1.8485]$$

$$= 0.9936$$

$$y^{(VI)} = \frac{1}{17}\left[35 - 2x^{(VI)} - 4z^{(V)}\right] = \frac{1}{17}[35 - 2\times 0.9936 - 4\times 1.8485]$$

$$= 1.5070$$

$$z^{(VI)} = \frac{1}{10}\left[24 - x^{(VI)} - 3y^{(VI)}\right] = \frac{1}{10}[24 - 0.9936 - 3\times 1.5070]$$

$$= 1.8485$$

Solution

$x = 0.9936$, $y = 1.5070$, $z = 1.8485$

Interpolation.

Interpolation means the process of computing intermediate values of a function from a given set of tabular value of the function

Four cases :- Newton's forward
" backward.
Lagranges method.
Newton's divided difference.

- If x values are equally spaced and interpolation at the beginning of table → Newton's forward
- If x values are equally spaced and interpolation at the end of the table → Newton's backward.

· Lagrange's and Newton's divided difference method for unequal difference

Forward difference table $[\Delta y_n = y_{n+1} - y_n]$

x	y	Δ	Δ^2	Δ^3	Δ^4
x_0	y_0	$y_1 - y_0 = \Delta y_0$	$\Delta y_1 - \Delta y_0 = \Delta^2 y_0$	$\Delta^2 y_1 - \Delta^2 y_0 = \Delta^3 y_0$	$\Delta^3 y_1 - \Delta^3 y_0 = \Delta^4 y_0$
x_1	y_1	$y_2 - y_1 = \Delta y_1$	$\Delta y_2 - \Delta y_1 = \Delta^2 y_1$	$\Delta^2 y_2 - \Delta^2 y_1 = \Delta^3 y_1$	
x_2	y_2	$y_3 - y_2 = \Delta y_2$	$\Delta y_3 - \Delta y_2 = \Delta^2 y_2$		
x_3	y_3	$y_4 - y_3 = \Delta y_3$			
x_4	y_4				

Backward difference table

x	y	∇	∇^2	∇^3	∇^4
x_0	y_0				
x_1	y_1	$y_1 - y_0 = \nabla y_1$	$\nabla y_2 - \nabla y_1 = \nabla^2 y_2$	$\nabla^2 y_3 - \nabla^2 y_2 = \nabla^3 y_3$	
x_2	y_2	$y_2 - y_1 = \nabla y_2$	$\nabla y_3 - \nabla y_2 = \nabla^2 y_3$	$\nabla^2 y_4 - \nabla^2 y_3 = \nabla^3 y_4$	$\nabla^3 y_4 - \nabla^3 y_3 = \nabla^4 y_4$
x_3	y_3	$y_3 - y_2 = \nabla y_3$	$\nabla y_4 - \nabla y_3 = \nabla^2 y_4$		
x_4	y_4	$y_4 - y_3 = \nabla y_4$			

Relation Between Operators

1) $\Delta = E - 1$
2) $\nabla = 1 - E^{-1}$
3) $\delta = E^{1/2} - E^{-1/2}$
4) $E = e^{hD}$ h = Interval, D = differentiation operator.

Newton's forward Interpolation formula

$x : x_0, x_1, x_2 \ldots x_n$

$y : y_0, y_1, y_2 \ldots y_n$

$x : x_0 + h, x_2 = x_1 + h = x_0 + 2h, x_3 = x_0 + 3h$

y at $x = x_0 + ph$ where p is small interval value ($e.g. \frac{1}{3}, \frac{1}{2} \ldots$)

$$f(x_0 + ph) = E^p f(x_0)$$
$$= (1 + \Delta)^p f(x_0)$$
$$= \left(1 + p\Delta + \frac{p(p-1)\Delta^2}{2!} + \cdots\right) f(x_0)$$

$Ef(x) = f(x+h)$
$E^2 f(x) = f(x+2h)$
$E^3 f(x) = f(x+3h)$
$\Delta = E - 1$

$$f(x_0 + ph) = f(x_0) + p\Delta f(x_0) + \frac{p(p-1)\Delta^2 f(x_0)}{2!} + \ldots$$

Newton's backward interpolation formula

$$f(x_n + ph) = f(x_n) + p\nabla f(x_n) + \frac{p(p+1)}{2!}\nabla^2 f(x_n) + \frac{p(p+1)(p+2)}{3!}\nabla^3 f(x_n) + \ldots$$

Derivation

$\nabla = 1 - E^{-1}$

$f(x_n + ph) = E^p f(x_n)$

$= (E^{-1})^{-p} f(x_n)$

$= (1 - \nabla)^{-p} f(x_n)$

$= \left(1 + p\nabla + \frac{p(p+1)}{2!}\nabla^2 + \dots\right) f(x_n)$

Q) Find the value of y at $x = 21$ and at $x = 28$ from the following table.

→ x	y	Δ	Δ^2	Δ^3
20	0.3420	0.0487		
23	0.3907		-1×10^{-3}	
		0.0477		-3×10^{-4}
26	0.4384		-1.3×10^{-3}	
		0.0464		
29	0.4848			

$h = 3$.

$x = 21$ → beginning → Forward interpolation

$x = 28$ → end → backward interpolation

$f(x_0 + ph) = f(x_0) + p\,\Delta f(x_0) + \frac{p(p-1)}{2!}\Delta^2 f(x_0) + \frac{p(p-1)(p-2)}{3!}\Delta^3 f(x_0)$

$x = 21 \quad x_0 = 20 \quad h = 3$

Forward interpolation $p = \frac{x - x_0}{h} = \frac{21 - 20}{3} = 0.33 = \frac{1}{3}$

$f(x_0) = 0.3420 \quad \Delta f(x_0) = 0.0487 \quad \Delta^2 f(x_0) = -1 \times 10^{-3}$

$\Delta^3 f(x_0) = -3 \times 10^{-4}$

$f(x_0 + ph) = 0.3420 + \frac{1}{3} \times 0.0487 + \frac{\frac{1}{3}\left(\frac{1}{3} - 1\right)}{2!} \times -1 \times 10^{-3}$

$+ \frac{\frac{1}{3}\left(\frac{1}{3} - 1\right)\left(\frac{1}{3} - 2\right)}{3!} \times -3 \times 10^{-4}$

$= 0.3420 + \frac{0.0487}{3} + \frac{-1}{9} \times -1 \times 10^{-3} + \frac{-1}{54000}$

$= 0.3583$

Backward interpolation

$f(x_n + ph) = f(x_n) + p\nabla(fx_n) + \frac{p(p+1)}{2!}\nabla^2 f(x_n) + \dots$

$x = 28, \quad x_n = 29, \quad p = \frac{x - x_n}{h} = \frac{28 - 29}{3} = \frac{-1}{3}$

$f(x_n) = 0.4848 \quad \nabla f(x_n) = 0.0464 \quad \nabla^2 f(x_n) = -1.3 \times 10^{-3}$

$\nabla^3 f(x_n) = -3 \times 10^{-4}$

$f(x_n + ph) = 0.4848 + \frac{-1}{3} \times 0.0464 + \frac{\frac{-1}{3}\left(\frac{-1}{3} + 1\right)}{2!} \times -1.3 \times 10^{-3}$

$+ \frac{\frac{-1}{3}\left(\frac{-1}{3} + 1\right)\left(\frac{-1}{3} + 2\right)}{3!} \times -3 \times 10^{-3}$

$= 0.4848 - \frac{0.0464}{3} + \frac{13}{90000} + \frac{1}{5400} = 0.46907$

Q) Find a polynomial of degree 2 which takes the following values.

x	y	Δ	Δ^2	Δ^3
0	1			
		1		
1	2		1	
		2		0
2	4		1	
		3		0
3	7		1	
		4		0
4	11		1	
		5		0
5	16		1	
		6		0
6	22		1	
		7		
7	29			

$$f(x_0+ph) = f(x_0) + p\Delta f(x_0) + \frac{p(p-1)}{2!}\Delta^2 f(x_0)$$

$$f(x_0) = 1 \qquad \Delta f(x_0) = 1 \qquad \Delta^2 f(x_0) = 1$$

$$p = \frac{x-x_0}{h} = \frac{x-0}{1} = x$$

$$f(x_0+ph) = 1 + x + \frac{x(x-1)}{2!} \times 1$$

$$= 1 + x + \frac{x(x-1)}{2}$$

$$= \frac{2+2x+x(x-1)}{2}$$

Q) Population of a town.

year (x)	Population in lakhs (y)	1Δ	Δ^2	Δ^3	Δ^4	Δ^5
1941	20					
		4				
1951	24		1			
		5		1		
1961	29		2		0	
		7		1		1
1971	36		3		1	
		10		2		
1981	46		5			
		5				
1991	51					

Estimate the population in year 1946 and 1976.

Backward interpolation

$$f(x_0+ph) = f(x_0) + p\Delta f(x_0) + \frac{p(p-1)}{2!}\Delta^2 f(x_0) + \frac{p(p-1)(p-2)}{3!}\Delta^3 f(x_0)$$

$$+ \frac{p(p-1)(p-2)(p-3)}{4!}\Delta^4 f(x_0) + \frac{p(p-1)(p-2)(p-3)(p-4)}{5!} \times \Delta^5 f(x_0)$$

$$p = \frac{x-x_0}{h} = \frac{1946-1941}{10} = 0.5$$

$$= 20 + 0.5 \times 4 + \frac{0.5(0.5-1)}{2!} \times 1 + \frac{0.5(0.5-1)(0.5-2)}{3!} \times 1$$

$$+ \frac{0.5(0.5-1)(0.5-2)(0.5-3)}{4!} \times 0 + \frac{0.5(0.5-1)(0.5-2)(0.5-3)(0.5-4)}{5!}$$

$$= 22.6211$$

Backward interpolation

$$f(x_n+ph) = f(x_n) + p\nabla f(x_n) + \frac{p(p+1)}{2!}\nabla^2 f(x_n) + \frac{p(p+1)(p+2)}{3!}\nabla^3 f(x_n) + \frac{p(p+1)(p+2)(p+3)}{4!}\nabla^4 f(x_n) + \frac{p(p+1)(p+2)(p+3)(p+4)}{5!}\nabla^5 f(x_n)$$

using x_n as 1984

$$p = \frac{x-x_n}{h} = \frac{1976-1981}{10} = -\frac{1}{2}$$

$$= 46 + \frac{-1}{2}\times 10 + \frac{\frac{-1}{2}\left(\frac{-1}{2}+1\right)}{2!}\times 10 + \frac{\frac{-1}{2}\left(\frac{-1}{2}+1\right)\left(\frac{-1}{2}+2\right)}{3!}\times$$

$$= 46 + \frac{-1}{2}\times 10 + \frac{\frac{-1}{2}\left(\frac{-1}{2}+1\right)}{2!}\times 3 + \frac{\frac{-1}{2}\left(\frac{-1}{2}+1\right)\left(\frac{-1}{2}+2\right)}{3!}\times 1$$

$$+ \frac{\frac{-1}{2}\left(\frac{-1}{2}+1\right)\left(\frac{-1}{2}+2\right)\left(\frac{-1}{2}+3\right)}{4!}\times 0 + \frac{\frac{-1}{2}\left(\frac{-1}{2}+1\right)\left(\frac{-1}{2}+2\right)\left(\frac{-1}{2}+3\right)\left(\frac{-1}{2}+4\right)}{5!}\times 1$$

$$= 46 + -5 + -0.375 + -0.0625 + +0.0273$$

$$= 40.53$$

using x_n as 1991

$$p = \frac{1976-1991}{10} = \frac{-3}{2}$$

$$= 46 + \frac{-3}{2}\times 5 + \frac{\frac{-3}{2}\left(\frac{-3}{2}+1\right)}{2!}\times 5 + \frac{\frac{-3}{2}\left(\frac{-3}{2}+1\right)\left(\frac{-3}{2}+2\right)}{3!}\times 2$$

$$+ \frac{\frac{-3}{2}\left(\frac{-3}{2}+1\right)\left(\frac{-3}{2}+2\right)\left(\frac{-3}{2}+3\right)}{4!}\times 1 + \frac{\frac{-3}{2}\left(\frac{-3}{2}+1\right)\left(\frac{-3}{2}+2\right)\left(\frac{-3}{2}+3\right)\left(\frac{-3}{2}+4\right)}{5!}\times 1$$

$$= 51 + -7.5 + \underset{+1.875}{-0.625} + \underset{+0.125}{-0.125} + \underset{+0.0234}{-0.039} + \underset{+0.01171}{-0.0273}$$

$$= 47.68 = 45.53$$

Lagrange's Formula

Let

x: $x_0\ x_1\ x_2 \ldots x_n$

y: $y_0\ y_1\ y_2 \ldots y_n$

$$y(x) = \frac{(x-x_1)(x-x_2)\ldots(x-x_n)}{(x_0-x_1)(x_0-x_2)\ldots(x_0-x_n)}y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)\ldots(x-x_n)}{(x_1-x_0)(x_1-x_2)\ldots(x_1-x_n)}y_1 + \ldots + \frac{(x-x_0)(x-x_1)\ldots(x-x_{n-1})}{(x_n-x_0)(x_n-x_1)\ldots(x_n-x_{n-1})}y_n$$

Q) Using Lagranges formula, find $y(10)$ from the following table

$x_0=5 \quad x_1=6 \quad x_2=9, \quad x_3=11$

$y_0=12 \quad y_1=13 \quad y_2=14, \quad y_3=16$

$x: 5, 6, 9, 11$

$y: 12, 13, 14, 16$

$$y(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} y_1 + \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} y_3$$

$$y(x) = \frac{(10-6)(10-9)(10-11)}{(5-6)(5-9)(5-11)} \times 12 + \frac{(10-5)(10-9)(10-11)}{(6-5)(6-9)(6-11)} \times 13 + \frac{(10-5)(10-6)(10-11)}{(9-5)(9-6)(9-11)} \times 14 + \frac{(10-5)(10-6)(10-9)}{(11-5)(11-6)(11-9)} \times$$

$$= 2 + \frac{-13}{3} + \frac{35}{3} + \frac{16}{3} = 14.66$$

Q) Use lagranges formula to fit a polynomial to given data.

	x_0	x_1	x_2	x_3
x:	-1	0	2	3
	y_0	y_1	y_2	y_3
y:	-8	3	1	12

And hence find $y(1)$

$$y(x) = \frac{(x-0)(x-2)(x-3)}{(-1-0)(-1-2)(-1-3)} \times -8 + \frac{(x+1)(x-2)(x-3)}{(0+1)(0-2)(0-3)} \times 3 + \frac{(x+1)(x-0)(x-3)}{(2+1)(2-0)(2-3)} \times 1 + \frac{(x+1)(x-0)(x-2)}{(3+1)(3-0)(3-2)} \times 12.$$

$$= \frac{-8x(x-2)(x-3)}{-12} + \frac{3(x+1)(x-2)(x-3)}{6} + \frac{(x+1)(x-3)x}{-6} + \frac{12x(x+1)(x-2)}{12}$$

$$= \frac{-8x(x-2)(x-3)}{-12} + \frac{12x(x+1)(x-2)}{12} + \frac{8(x+1)(x-2)(x-3)}{8} + \frac{(x+1)(x-3)}{-6}$$

$$= \frac{2}{3}(x^2-3x-2x+6) + x[x^2-2x+x-2] + \frac{1}{2}[x^2-2x+x-2](x-3) + x^2 - 3x + x - 3$$

$= \frac{2}{3}[x^2-5x+6] = \frac{2}{3}[x^3-5x^2+6x] + (x^3-2x^2+x^2-2x)$

$+\frac{1}{2}[x^3-2x^2+x^2-2x-3x^2+6x-3x+6]$

$-\frac{1}{6}[x^2-2x+3]$

$= \frac{2}{3}[x^3-5x^2+6x] + [x^3-x^2-2x] + \frac{1}{2}[x^3-4x^2+x+6]$

$-\frac{1}{6}[x^2-2x+3]$

$= 2x^3-6x^2+3x+3$

$y(1) = \underline{2}$

Newton's divided difference formula.

x	y	Δ	Δ^2	Δ^3
x_0	y_0			
		$\frac{y_1-y_0}{x_1-x_0} = [x_0,x_1]$		
x_1	y_1		$\frac{[x_1,x_2]-[x_0,x_1]}{x_2-x_0} = [x_0,x_1,x_2]$	
		$\frac{y_2-y_1}{x_2-x_1} = [x_1,x_2]$		$\frac{[x_2,x_3]-[x_1,x_2]}{x_3-x_1} = [x_1,x_2,x_3]$
x_2	y_2			
		$\frac{y_3-y_2}{x_3-x_2} = [x_2,x_3]$		
x_3	y_3			

Δ^3

$\frac{[x_1,x_2,x_3][x_0,x_1,x_2]}{x_3-x_0} = [x_0,x_1,x_2,x_3]$

Q) Find divided differences of $f(x) = x^3+x+2$ for the arguments 1, 3, 6, 11

x	$y=f(x)$	Δ	Δ^2	Δ^3
1	4	$\frac{32-4}{3-1} = 14$		
3	32		$\frac{64-14}{6-1} = 10$	
		$\frac{224-32}{6-3} = 64$		$\frac{20-10}{11-1} = 1$
6	224		$\frac{224-64}{11-3} = 20$	
		$\frac{1344-224}{11-6} = 224$		
11	1344			

→ Newton's divided difference Formula

Formula

$f(x) = f(x_0) + (x-x_0)\times[x_0x_1] + (x-x_0)(x-x_1)[x_0x_1x_2]$

$+ \ldots\ldots + (x-x_0)(x-x_1)\cdots[x-x_{n-1}]$

$[x_0,x_1,x_2\ldots x_n]$

Q) From the following table, find $f(8)$ using newton's divided different formal

x	$f(x)$	Δ	Δ^2	Δ^3	Δ^4	Δ^5
x_0 4	48					
		$\frac{100-48}{5-4} = 52 = [x_0x_1]$				
x_1 5	100		$\frac{294-100}{7-5} = 97 = [x_1x_2]$			
		$202 = [x_2,x_3]$		$\frac{202-97}{7-4} = 35 = [x_0,x_1,x_2,x_3]$		
x_2 7	294		$\frac{3010-202}{10-5} = 561.6 = [x_1,x_2,x_3,x_4]$		$\frac{561.6-35}{8-4} = 131.65 = [x_0,x_1,x_2,x_3,x_4]$	
		$3010 = [x_3,x_4]$		$\frac{3010-561.6}{10-5} = 489.68 = [x_2,x_3,x_4,x_5]$		$\frac{489.68-131.65}{8-4} = 89.5075 = [x_0,x_1,x_2,x_3,x_4,x_5]$
x_3 10	900		$\frac{1210-3010}{11-7} = 450 = [x_3,x_4,x_5]$			
		$409 = [x_4,x_5]$				
x_4 11	1210					
x_5 13	2028					

$f(8) = 48 + (8-4)\times[52] + [8-4][8-5][15]$

$+ [8-4][8-5][8-7]\times 1 + 0 + 0$

$= \underline{448}$

2) Find a polynomial corresponding to the given data using Newton's divided difference formula and hence find y at $x=0$

	x	y	Δ	Δ^2	Δ^3
x_0	-1	-21			
			18 $[x_0,x_1]$		
x_1	1	15		-7 $[x_0,x_1,x_2]$	
			-3 $[x_1,x_2]$		1 $[x_0,x_1,x_2,x_3]$
x_2	2	12		-3 $[x_1,x_2,x_3]$	
			-9 $[x_2,x_3]$		
x_3	3	3			

$$f(x) = f(x_0) + (x-x_0)[x_0,x_1] + (x-x_0)(x-x_1)[x_0,x_1,x_2] + (x-x_0)(x-x_1)(x-x_2)[x_0,x_1,x_2,x_3]$$

$$f(0) = -21 + (x--1)\times 18 + (x--1)(x-2)\times -7 + (x--1)(x-1)(x-2)\times 1$$

$$= -21 + (x+1)18 + (x+1)(x-1)x-7 + (x+1)(x-1)(x-2)$$

$$= -21 + 18x + 18 + (x^2-1)x - 7 + (x^2-1)(x-2)$$

$$= -21 + 18x + 18 - 7x^2 + 7 + x^3 - 2x^2 - x + 2$$

$$= x^3 - 9x^2 + 17x + 6$$

$$f(0) = 6$$

Method of least Squares

Fitting of straight line

Let $y = ax+b$, then normal equations are :-

$$\Sigma y = a\Sigma x + nb$$

$$\Sigma xy = a\Sigma x^2 + b\Sigma x$$

- if x values are $x_1, x_2, x_3 \cdots x_n$ & y values are $y_1, y_2, y_3 \cdots y_n$
- Find Σx, Σy, Σx^2, Σxy from the given values and substitute in normal equation.
- By solving normal equation, we can find a and b.

Fitting of Quadratic equation

→ Let $y = ax^2+bx+c$, then normal equations are

$$\Sigma y = a\Sigma x^2 + b\Sigma x + nc$$

$$\Sigma xy = a\Sigma x^3 + b\Sigma x^2 + c\Sigma x$$

$$\Sigma x^2 y = a\Sigma x^4 + b\Sigma x^3 + c\Sigma x^2$$

Fitting of exponential curve and power curve

I $y = ax^b$

$\log y = \log a + b\log x$

$Y = A + bX$

~~$\Sigma Y = a\Sigma X + n$~~

$\Sigma Y = nA + b\Sigma X$

$\Sigma XY = A\Sigma X + b\Sigma X^2$

II $y = ab^x$

$\log y = \log a + x\log b$

$Y = A + Bx$

$\Sigma Y = nA + B\Sigma x$

$\Sigma XY = A\Sigma x + B\Sigma x^2$

III $y = ae^{bx}$

$\log y = \log a + bx\log e$.

$Y = A + Bx$.

$\Sigma Y = nA + B\Sigma x$

$\Sigma XY = A\Sigma x + B\Sigma x^2$

Q) Fit a straight line to the given data. Also estimate the value of y at $x = 2.5$.

x	y
0	1
1	1.8
2	3.3
3	4.5
4	6.3

General form $y = ax + b$.

Normal equations are $\Sigma y = a\Sigma x + nb$

$\Sigma xy = a\Sigma x^2 + b\Sigma x$

$\Sigma x = 10$

$\Sigma y = 16.9$

$\Sigma x^2 = 30$

$\Sigma xy = 47.1$

$n = 5$

$16.9 = a \times 10 + 5 \times b$.

$16.9 = 10a + 5b$ — ①.

$47.1 = a \times 30 + b \times 10$

$47.1 = 30a + 10b$ — ②

$a = 1.33$, $b = 0.72$

$y = 1.33x + 0.72$

$y(2.5) = 1.33 \times 2.5 + 0.72$

$= 4.045$

Q) Fit a Parabola by method of least squares to the following data

→

x	y
1	5
2	12
3	26
4	60
5	97.

General form $y = ax^2 + bx + c$.

$\Sigma y = a\Sigma x^2 + b\Sigma x + nc$

$\Sigma xy = a\Sigma x^3 + b\Sigma x^2 + c\Sigma x$

$\Sigma x^2 y = a\Sigma x^4 + b\Sigma x^3 + c\Sigma x^2$

$\Sigma x = 15$ $\quad \Sigma y = 200$.

$\Sigma x^2 = 55$. $\quad \Sigma xy = 832$.

$\Sigma x^3 = 225$. $\quad \Sigma x^2 y = 3672$

$\Sigma x^4 = 979$

$\Sigma y = a \times 55 + b \times 15 + 5c$.

$200 = 55a + 15b + 5c$.

$832 = 225a + 55b + 15c$.

$3672 = 979a + 225b + 55c$.

$a = 10.49$ $b = -25.04$ $c = -0.26$

$a = 5.71$, $b = -11.0$, $c = 10.4$

$y = 5.71x^2 - 11x + 10.4$

Q) Fit a curve of the form $y = ab^x$ to the following data

→

x	y
1	151
2	100
3	61
4	50
5	20
6	8

$y = ab^x$.

$\log y = \log a + x \log b$.

$Y = A + Bx$

normal eq^n

$\Sigma Y = nA + B\Sigma x$.

$\Sigma xY = A\Sigma x + B\Sigma x^2$

$\Sigma Y = 9.8673 \quad \Sigma x = 21$

$\Sigma xY = 30.2545 \quad \Sigma x^2 = 91$

$9.8673 = 6A + 21B$

$30.2545 = 21A + 91B$

$A = 2.5 \quad B = -0.24$

$Y = 2.5 - 0.24x$

$\log a = 2.5 \quad \log b = -0.24$

$a = 316.22 \quad b = 0.575$.

$y = 316.2 \times 0.575^x$

Q) From the table given below. Find the best values of a and b. $y = ae^{bx}$ by the method of least squares.

x	y
0	3
5	1.5
8	1
12	0.55
20	0.18

$y = ae^{bx}$ $\qquad B = b \times \log e$.

$\log y = \log a + bx \log e$.

$Y = A + Bx$

$\Sigma Y = nA + B\Sigma x$

$\Sigma xY = A\Sigma x + B\Sigma x^2$ 10

$\Sigma Y = \Sigma \log y$

$\Sigma Y = -0.35$.

$\Sigma x = 45 \qquad \Sigma x^2 = 633$

$\Sigma xY = \Sigma x \times \log y$

$= -17.13$

$-0.35 = 5A + 45B$

$-17.13 = 45A + 633B$

$A = 0.48 \quad B = -0.061$

$a = 3.02 \quad B = b \times \log e$

$b = \frac{B}{\log e} = -0.14$

$a = 10^{0.48}$

Q) Find the best values of a and b in $y = ax^b$ by the method of least squares from the given data

x	y
1	1200
2	900
3	600
4	200
5	110
6	50

$y = ax^b$

$\log y = \log a + b \log x$

$Y = A + bX$

$\Sigma Y = nA + b\Sigma X$

$\Sigma XY = A\Sigma X + b\Sigma X^2$

$A\Sigma X^2 + b\Sigma X$

$3060 = 6A + 21b$

$6450 = 21A + 91b$

$\Sigma Y = 3060 \quad \Sigma X = 21 \quad n = 6$

$\Sigma XY = 6450 \quad \Sigma X^2 = 91.$

$3060 = 6A + 21b$

$6450 = 21A + 91b$

$A = 1362 \quad b = -243.42$

$A = \log a$

$a = 10^{1362} =$

$\Sigma X = 2.857$

$\Sigma Y = 14.85.$

$\Sigma X^2 = 1.774$

$\Sigma XY = \Sigma \log x \cdot \log y$

$= 6.35$

$14.85 = 6A + 2.86b$

$6.85 = 2.86A + 1.77b$

$A = 3.33 \quad b = -1.79$

$A = 3.33$

$a = 10^{3.33} = 2137.96$

$a = 2137.96$

$b = -1.79$

Module-2.

Numerical Differentiation

1) Derivative using Newton's Forward Interpolation Formula

Newton $y = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \cdots$

$$\frac{dy}{dx} = \frac{dy}{dp}\cdot\frac{dp}{dx} = \frac{1}{h}\left[\Delta y_0 + \frac{2p-1}{2!}\Delta^2 y_0 + \frac{3p^2-6p+2}{3!}\Delta^3 y_0 + \cdots\right]$$

$$\frac{d^2y}{dx^2} = \frac{1}{h^2}\left[\Delta^2 y_0 + \frac{6p-6}{3!}\Delta^3 y_0 + \cdots\right]$$

Special case when $p=0$

when $p=0$,

$$\frac{dy}{dx} = \frac{1}{h}\left[\Delta y_0 - \frac{\Delta^2 y_0}{2} + \frac{\Delta^3 y_0}{3} - \frac{\Delta^4 y_0}{4} \cdots\right]$$

$$\frac{d^2y}{dx^2} = \frac{1}{h^2}\left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12}\Delta^4 y_0 - \cdots\right]$$

$p = \frac{x - x_0}{h}$

$\frac{dp}{dx} = \frac{1}{h}$

$\frac{d^2y}{dx^2} = \frac{dy}{dp}\left[\frac{dy}{dx}\right]$

2) Derivative using Newton's Backward interpolation Formula

$$y = y_n + p\nabla y_n + \frac{p(p+1)}{2!}\nabla^2 y_n + \frac{p(p+1)(p+2)}{3!}\nabla^3 y_n + \cdots$$

$$\frac{dy}{dx} = \frac{dy}{dp}\cdot\frac{dp}{dx} = \frac{1}{h}\left[\nabla y_n + \frac{2p+1}{2!}\nabla^2 y_n + \frac{3p^2+6p+2}{3!}\nabla^3 y_n + \cdots\right]$$

$$\frac{d^2y}{dx^2} = \frac{1}{h^2}\left[\nabla^2 y_n + \frac{6p+6}{3!}\nabla^3 y_n + \cdots\right]$$

when $p=0$

$$\frac{dy}{dx} = \frac{1}{h}\left[\nabla y_n + \frac{\nabla^2 y_n}{2} + \frac{\nabla^3 y_n}{3} + \cdots\right]$$

$$\frac{d^2y}{dx^2} = \frac{1}{h^2}\left[\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12}\nabla^4 y_n + \cdots\right]$$

$\frac{P(P-1)(P-2)(P-3)}{4!}\Delta^4 y_0$

Derivatives using central Defense Difference

p^3-p
p^4-p^2

* ~~Sterli~~ Starli
* Stirling's Formula

$$y = y_0 + \frac{p}{2}(\Delta y_0 + \Delta y_{-1}) + \frac{p^2}{2}\Delta^2 y_{-1} + \frac{p(p^2-1)}{3!}\left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2}\right) + \frac{p^2(p^2-1)}{4!}\Delta^4 y_{-2} + \dots$$

$$\frac{dy}{dx} = \frac{dy}{dp}\frac{dp}{dx} = \frac{1}{h}\left[\frac{\Delta y_0 + \Delta y_{-1}}{2} + p\Delta^2 y_{-1} + \frac{3p^2-1}{3!}\left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2}\right) + \frac{4p^3-2p}{4!}\Delta^4 y_{-2} + \dots\right]$$

$$\frac{d^2y}{dx^2} = \frac{1}{h^2}\left[\Delta^2 y_{-1} + \frac{6p}{3!}\left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2}\right) + \frac{12p^2-2}{4!}\Delta^4 y_{-2} + \dots\right]$$

when $p=0$

$$\frac{dy}{dx} = \frac{1}{h}\left[\frac{\Delta y_0 + \Delta y_{-1}}{2} - \frac{1}{12}(\Delta^3 y_{-1} + \Delta^3 y_{-2}) + \dots\right]$$

$$\frac{d^2y}{dx^2} = \frac{1}{h^2}\left[\Delta^2 y_{-1} - \frac{\Delta^4 y_{-2}}{12} + \dots\right]$$

Q) Find the 1st and 2nd derivatives of y at $x=50$, and $x=56$ from the given table.

x	y	Δy	Δy^2	Δy^3
50	3.684			
51	3.7084	0.0244	-3×10^{-4}	
52	3.7325	0.0241	-3×10^{-4}	0
53	3.7563	0.0238	-3×10^{-4}	0
54	3.7798	0.0235	-3×10^{-4}	0
55	3.8030	0.0232	-3×10^{-4}	0
56	3.8259	0.0229		

$h=1$

$x=50$, beginning of the table. $x=50=x_0$ $x=x_0$, $p=0$
$x=56$, end of table. $x=56=x_n$ $x=x_n$, $p=0$

when $p=0$ [Forward]. at $x=50$

$$\frac{dy}{dx} = \frac{1}{h}\left[\Delta y_0 - \frac{\Delta^2 y_0}{2}\right]$$

$$= 1\left[0.0244 - \frac{-3\times10^{-4}}{2}\right] = 0.02455$$

$$\frac{d^2y}{dx^2} = \frac{1}{h^2}\left[\Delta^2 y_0\right] = \frac{1}{1}\times -3\times10^{-4} = -3\times10^{-4}$$

when $p=0$ [backward] at $x=56$.

$$\frac{dy}{dx} = \frac{1}{h}\left[\nabla y_n + \frac{\nabla^2 y_n}{2}\right] = 1\left[0.0229 + \frac{-3\times10^{-4}}{2}\right]$$

$$= 0.02275$$

$$\frac{d^2y}{dx^2} = \frac{1}{h^2}\left[\nabla^2 y_n\right] = \frac{1}{1}\times -3\times10^{-4} = -3\times10^{-4}$$

Q The population of a town is given below. Find the rate of growth of population in 1931, 1941 and 1971 from the given table

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1931	40.62	20.18			
1941	60.80		-1.03		
1951	79.95	19.15		5.49	-4.47
1961	103.56	23.61	4.46		
1971	132.65	29.09	5.48	1.02	

At $x = 1931$, $p = 0$, Forward interpolation

$$\frac{dy}{dx} = \frac{1}{h}\left[\Delta y_0 - \frac{\Delta^2 y_0}{2} + \frac{\Delta^3 y_0}{3} - \frac{\Delta^4 y_0}{4}\right]$$

$$= \frac{1}{10}\left[40.62 - \frac{20.18}{2} + \frac{1.03}{3} - \right.$$

$$= \frac{1}{10}\left[20.18 + \frac{1.03}{2} + \frac{5.49}{3} + \frac{4.47}{4}\right]$$

$$= 2.36$$

At $x = 1971$, $p = 0$, Backward interpolation

$$\frac{dy}{dx} = \frac{1}{h}\left[\nabla y_n + \frac{\nabla^2 y_n}{2} + \frac{\nabla^3 y_n}{3} + \frac{\nabla^4 y_n}{4}\right]$$

$$= \frac{1}{10}\left[29.09 + \frac{5.48}{2} + \frac{1.02}{3} + \frac{-4.47}{4}\right]$$

$$= 3.105$$

At $x = 1941$

$$\frac{dy}{dx} = \frac{1}{h}\left[\Delta y_0 + \frac{2p-1}{2!}\Delta^2 y_0 + \frac{3p^2-6p+2}{3!}\Delta^3 y_0 + \frac{4p^3-18p^2+22p-6}{4!}\Delta^4 y_0\right]$$

$$= \frac{1}{10}\left[20.18 + \frac{2-1}{2} \times -1.03 + \frac{3-6+2}{3!} \times 5.49 + \frac{4-18+22-6}{4!} \times 4.47\right]$$

$$= \frac{1}{10}\left[20.18 + \frac{-1.03}{2} + \frac{-1}{3!} \times 5.49 + \frac{2}{4!} \times -4.47\right]$$

$$= 1.838$$

Q) Find the gradient of the road at the middle point of the elevation about a line of 7 points of road which are given below

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$	$\Delta^6 y$
x=300	135	14					
x=600	149	80	-6	24			
x=900	157	26	18	-26	-50	70	-86
	183	18	-8	-6	20	-16	
1200	201		-14		4		
1500	205	4	-16	-2			
1800	193	-12					

At $x = 900$, $p = \frac{x - x_0}{h}$ $p = 0$ $x_0 = 900$. $y_0 = 183$.

When $p = 0$

$$\frac{dy}{dx} = \frac{1}{h}\left[\frac{\Delta y_0 + \Delta y_{-1}}{2} - \frac{1}{12}\left(\Delta^3 y_{-1} + \Delta^3 y_{-2}\right) + \frac{1}{60}\left(\Delta^5 y_{-2} + \Delta^5 y_{-3}\right)\right.$$

$$\frac{p(p-1)(p-2)(p-3)\Delta^4 y}{4!}$$

$$\frac{p(p-1)(p^2-3p-2p+6)}{4!}$$

$$\frac{p(p-1)(p^2-5p+6)}{4!}$$

$$p[p^3-5p^2+6p - p^2+5p-6]$$

$$p^4-5p^3+6p^2 - p^3+5p^2-6p$$

$$p^4-6p^3+11p^2-6p$$

$$4p^3-18p^2+22p-6$$

$$p = \frac{[x_0 - x_n]}{2h}$$

$$\frac{x_n - x_0}{2h} = \frac{10}{10} = 1$$

$= \frac{1}{300}\left[\frac{18+26}{2} - \frac{1}{12}\left[6+(-26)\right] + \frac{1}{60}\left(-6+70\right)\right]$

$= 0.08052$

H.W.

Q) Find the 1st and 2nd derivative of the function tabulated below at $x = 0.6$.

	x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
x_{-2}	0.4	1.5836	0.2128	0.0330		
x_{-1}	0.5	1.7974	0.2468		3.5×10^{-3}	
x_0	0.6	2.0442		0.0865		0.3×10^{-3}
	0.7	2.3275	0.2833		3.8×10^{-3}	
	0.8	2.6511	0.3236	0.0403		

Q) The table given below is the velocity of a body during the time t specified. Find its acceleration at $t = 1.1$

t	v
1	43.1
1.1	47.7
1.2	52.1
1.3	56.4
1.4	60.8

Q1) At $x = 0.6$, middle point $p = 0$.

$$\frac{dy}{dx} = \frac{1}{h}\left[\frac{\Delta y_0 + \Delta y_{-1}}{2} - \frac{1}{12}\left[\Delta^3 y_{-1} + \Delta^3 y_{-2}\right]\right]$$

$$= \frac{1}{0.1}\left[\frac{0.2833 + 0.2468}{2} - \frac{1}{12}\left[3.8\times10^{-3} + 3.5\times10^{-3}\right]\right]$$

$$= 2.6544$$

$$\frac{d^2y}{dx^2} = \frac{1}{h^2}\left[\Delta^2 y_{-1} - \frac{\Delta^4 y_{-2}}{12}\right]$$

$$= \frac{1}{0.1^2}\left[0.0365 - \frac{0.3\times10^{-3}}{12}\right] = 3.6475$$

Q2)

t	v	Δv	$\Delta^2 v$	$\Delta^3 v$	$\Delta^4 v$
1	43.1				
1.1	47.7	4.6			
1.2	52.1	4.4	-0.2		
1.3	56.4	4.3	-0.1	0.1	
1.4	60.8	4.4	0.1	0.2	0.1

$h = 0.1 \quad p = \frac{x - x_0}{h} = \frac{1.1-1}{0.1} = 1$

Forward Interpolation

$$v = v_0 + p\Delta v_0 + \frac{p(p-1)}{2!}\Delta^2 v_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 v_0 + \frac{p(p-1)(p-2)(p-3)}{4!}\Delta^4 v_0$$

$$\frac{dv}{dt} = \frac{dv}{dp}\cdot\frac{dp}{dx} = \frac{1}{h}\left[\Delta v_0 + \frac{2p-1}{2!}\Delta^2 v_0 + \frac{3p^2-6p+2}{3!}\Delta^3 v_0 + \frac{4p^3-18p^2+22p-6}{4!}\Delta^4 v_0\right]$$

$$= \frac{1}{0.1}\left[4.6 + \frac{2\times1-1}{2!}\times0.2 + \frac{3-6+2}{3!}\times0.1 + \frac{4\times1-18+22-6}{4!}\times0.1\right]$$

$= 44.9167$

$\frac{dy}{dt} = v$

$dy = v\times dt$

$= 43.1\times$

$p = \frac{x - x_0}{h} = \frac{1.1-1}{0.1} = 1$

$\frac{dp}{dx}\cdot\frac{dy}{dp}$

$\frac{11\times0.1}{3\times2}$

Numerical Integration

I^st Method - Trapezoidal ~~Method~~ rule

$$\int_a^b f(x)\,dx = \frac{h}{2}\left[(y_0+y_n) + 2(y_1+y_2+\cdots+y_{n-1})\right]$$

SIMPSON'S 1/3 Rule

$$\int_a^b f(x)\,dx = \frac{h}{3}\left[(y_0+y_n) + 2(y_2+y_4+\cdots+y_{n-2}) + 4(y_1+y_3+\cdots+y_{n-1})\right]$$

SIMPSON'S 3/8 Rule

multiples of 3

$$\int_a^b f(x)\,dx = \frac{3h}{8}\left[(y_0+y_n) + 3(y_1+y_2+y_4+\cdots+y_{n-1}) + 2(y_3+y_6+\cdots+y_{n-3})\right]$$

* In Simpson's 1/3rd rule → no of sub intervals should be ~~equal~~ even
* In Simpson's 3/8th rule → no of sub intervals should be multiple of 3
* Trapezoidal rule → no of sub intervals → any number
* $h = \frac{b-a}{n}$, b = upper limit
  a = lower limit
  n = number of intervals
* If h is small, calculation complex
* If h is high, error more.
* h → appropriate value.

n = even (Simpson's 1/3 rule).

n = multiple of 3 (Simpson's 3/8)

If n is even & multiple of 3 ⇒ Simpson's 3/8.

Simpson's 3/8 ⇒ more accurate

Q) Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using trapezoidal rule

Here

$h = 0.2$ $\quad n = 5$

	x_0	x_1	x_2	x_3	x_4	x_5
x:	0,	0.2,	0.4,	0.6,	0.8	1
y:	1	0.9615	0.8621	~~0.7804~~ 0.7353,	0.6098	0.5

$\frac{1}{1+0}$ $\quad \frac{1}{1+0.2^2}$

$\int_0^1 \frac{dx}{1+x^2} = \frac{0.2}{2}$ [ ]

$\int_0^1 f(x)dx = \frac{h}{a}\left[(y_0+y_n)+2(y_1+y_2+\ldots y_{n-1})\right]$

$= \frac{0.2}{2}\left[(1+0.5)+2(0.9615+0.8621+0.7353+0.6098)\right]$

$= 0.7837$

Q) Evaluate $\int_4^{5.2} \log_e x\,dx$ using

(i) Trapezoidal rule

(ii) Simpson's $1/3^{rd}$ rule.

(iii) Simpson's $3/8^{th}$ rule.

n = multiple of 3 and also even.

(i)

$\Rightarrow h = 0.2$.

$n = 6$

x	4	4.2	4.4	4.6	4.8	5.0	5.2
y	1.3863 y_0	1.4351 y_1	1.4816 y_2	1.5261 y_3	1.5686 y_4	1.6094 y_5	1.6487 y_6

$h = \frac{b-a}{n} = \frac{5.2-4}{6} = 0.2$

$\int_4^{5.2} \log_e x\,dx = \frac{0.2}{2}\left[(1.3863+1.6487)+2(1.4351+1.4816+1.5261+1.5686+1.6094)\right]$

$= 1.8277$

(ii) Simpson's $1/3^{rd}$ rule

$\int_a^b f(x)dx = \frac{h}{3}\left[(y_0+y_n)+2(y_2+y_4+\ldots y_{n-2})+4(y_1+y_3+\ldots y_{n-1})\right]$

$\int_4^{5.2} \log_e x\,dx = \frac{0.2}{3}\left[(1.3863+1.6487)+2\left[1.4816+1.5686\right]+4\left[1.4351+1.5261+1.6094\right]\right]$

$= 1.8279$

(iii) Simpson's $2/3^{rd}$ rule

$\int_a^b f(x)dx = \frac{3h}{8}\left[(y_0+y_n)+3(y_1+y_2+y_4+\ldots+y_{n-1})+2(y_3+y_6+\ldots+y_{n-3})\right]$

$= \frac{3\times 0.2}{8}\left[(1.3863+1.6487)+3(1.4351+1.4816+1.5686+1.6094)+2(1.5261)\right]$

$= 1.8278$

Q) Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using $\int_0^1 \frac{dx}{1+x^2} = \frac{\pi}{4}$. $(\tan^{-1}x)_0^1$

(i) Simpson's $\frac{1}{3}$ rule taking $h = \frac{1}{4}$.

(ii) Simpson's $\frac{3}{8}$ rule taking $h = \frac{1}{6}$

Hence compute an approximate value of π in each case.

(i) $f(x) = \frac{1}{1+x^2}$ at $x = 0, \frac{1}{4}, \frac{2}{4}, \frac{3}{4}, 1$ are given below.

x	0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	1
$f(x)$	1	0.9412	0.8000	0.6400	0.5000
	y_0	y_1	y_2	y_3	y_4

Simpon's 1/3rd rule.

$$\int_a^b f(x)dx = \frac{h}{3}\left[(y_0+y_n) + 2\left[y_2+y_4+\cdots y_{n-2}\right] + 4\left[y_1+y_3+\cdots y_{n-1}\right]\right]$$

$$\int_0^1 \frac{dx}{1+x^2} = \frac{1/4}{3}\left[(1+0.5000) + 2\left[0.8000\right] + 4\left[0.9412+0.6400\right]\right]$$

$$= 0.7854$$

$$0.7854 = \frac{\pi}{4} \qquad \pi = \underline{\underline{3.1416}}$$

(ii) $\int_a^b f(x)dx = \frac{3h}{8}\left[(y_0+y_n) + 3\left[y_1+y_2+y_4+y_5+\cdots\right]\right]$

x	x_0	x_1	x_2	x_3	x_4	x_5	x_6
x	0	$\frac{1}{6}$	$\frac{2}{6}$	$\frac{3}{6}$	$\frac{4}{6}$	$\frac{5}{6}$	1
y	1	0.9730	0.9000	0.8000	0.6923	0.5902	0.5000
	y_0	y_1	y_2	y_3	y_4	y_5	y_6

Simpson's $\frac{3}{8}$ rule.

$$\int_a^b f(x)dx = \frac{3h}{8}\left[(y_0+y_6) + 3\left[y_1+y_2+y_4+y_5\right] + 2\left[y_3\right]\right]$$

$$= \frac{3\times 1}{8\times 6}\left[(1+0.5000) + 3\left[0.9730+0.9000+0.6923+0.5902\right] + 2\left[0.8000\right]\right]$$

$$= \underline{0.7854}$$

$$\int_0^1 \frac{dx}{1+x^2} = \frac{\pi}{4} = 0.7854$$

$$\pi = \underline{\underline{3.1416}}$$

Q) The speed, v metres per second of a car, t seconds after it starts, is shown in the following table. Simpson's $\frac{1}{3}$rd rule because $n=10$

using Simpson's rule, find the distance travelled by the car in 2min

t	0	12	24	36	48	60	72	84	96	108	120
v	0	3.60	10.08	18.90	21.60	18.54	10.26	5.40	4.50	5.40	9.00
	y_0	y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8	y_9	y_{10}

$$\frac{ds}{dt} = v$$

$$s = \int_0^{120} v\,dt$$

$$= \frac{h}{3}\left[(y_0+y_{10}) + 2[y_2+y_4+y_6+y_8] + 4[y_1+y_3+y_5+y_7+y_9]\right]$$

$$= \frac{12}{3}\left[(0+9.00) + 2[10.08+21.60+18.90+10.26] + 4[3.66+10.08+18.54+5.40+5.40]\right].$$

$$= 1236.96 \text{ m}$$

Q) A curve passes through the points $(1,2)$, $(1.5, 2.4)$, $(2.0, 2.7)$, $(2.5, 2.8)$, $(3, 3)$, $(3.5, 2.6)$ and $(4.0, 2.1)$. Obtain the area bounded by the curve, the x axis and $x=1$, $x=4$. Also find the volume of solid of revolution got by revolving this area about the x axis.

→ Area $= \int_a^b y\,dx = \int_1^4 y\,dx$ here $h = 0.5$. $n = 6$.

$$= \frac{h}{3}\left[(y_0+y_n) + 2[y_2+y_4] + 4[y_1+y_3+y_5]\right]$$

$$= \frac{0.5}{3}\left[(2+2.1) + 2[2.7+3] + 4[2.4+2.8+2.6]\right]$$

$$= 7.7834 \text{ sq. units}$$

Slope ⇒ differentiate.

Volume $= \pi \int_a^b y^2\,dx$.

$$\int_1^4 y^2\,dx = \frac{0.5}{3}\left[[2^2+(2.1)^2] + 2[(2.7)^2+3^2] + 4[2.4^2+2.8^2+2.6^2]\right]$$

$$= 20.405$$

Volume $= \pi \times 20.405 = 64.13$ cu. units

Two points Gaussian quadrature Formula

$$\int_{-1}^{1} f(x)\,dx = F\left(\frac{-1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right)$$

This formula is exact for polynomials upto degree 3

Three points Gaussian quadrature:

$$\int_{-1}^{1} f(x)\,dx = \frac{5}{9}\left[f(-\sqrt{0.6}) + f(\sqrt{0.6})\right] + \frac{8}{9}f(0)$$

General form

$$\int_a^b f(x)\,dx = \frac{b-a}{2}\int_{-1}^{1} f\left[\frac{b-a}{2}t + \frac{(b+a)}{2}\right]dt$$

$$= \frac{b-a}{2}\int_{-1}^{1} \phi(z)\,dz$$

Q) Apply Gauss two point formula to evaluate

(i) $\int_{-1}^{1} \frac{1}{1+x^2}\,dx$ (ii) $\int_0^1 \frac{dx}{1+x^2}$

$$\int_0^1 \frac{dx}{1+x^2} = \frac{1}{a}\int_{-1}^{1} \frac{dx}{1+x^2}$$

$$= \frac{1}{a} \times 1.5$$

$$= 0.7$$

(i) $f(x) = \frac{1}{1+x^2}$; $f\left(\frac{1}{\sqrt{3}}\right) = f\left(\frac{-1}{\sqrt{3}}\right) = \frac{3}{4}$.

$$\int_{-1}^{1} \frac{1}{1+x^2}\,dx = f\left(\frac{-1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right) = \frac{3}{4}+\frac{3}{4} = \frac{6}{4} = \frac{3}{2} = 1.5$$

$$\int_{-1}^{1} \frac{1}{1+x^2} = 2(\tan^{-1}x)_0^1 = \frac{\pi}{2} = 1.5708$$

even function.

$\int_{-1}^{1} f(x)\,dx$ even $= 2\int_0^1 f(x)\,dx$

2) using Gaussian 3 point Formula Evaluate

(i) $\int_{-1}^{1}(3x^2+5x^4)\,dx$ (ii) $\int_{0}^{1} 3x^2+5x^4\,dx$.

(i) $f(x) = 3x^2+5x^4$.

$f(\sqrt{0.6}) = f(\sqrt{0.6}) = 3(0.6)^2+5(0.6)^4$

$= 3\times0.6 + 5\times(0.6)^2$

$f(\sqrt{0.6}) = 3.6 \qquad f(0)=0$

$$\int_{-1}^{1} f(x)\,dx = \frac{5}{9}\left[f(-\sqrt{0.6})+f(\sqrt{0.6})\right]+\frac{8}{9}f(0)$$

$$= \frac{5}{9}\left[3.6+3.6\right]+\frac{8}{9}\times 0$$

$$= 4$$

(ii) $\int_{0}^{1} 3x^2+5x^4\,dx$

$= \frac{1}{2}\int_{-1}^{1}(3x^2+5x^4)\,dx = \frac{1}{2}\times 4 = 2$

3) Use Gaussian 3-point formula and evaluate.

$$I = \int_{1}^{5}\frac{dz}{z}.$$

⇒ put $z = \frac{b-a}{2}x + \frac{b+a}{2}$

$= \frac{5-1}{2}x + \frac{5+1}{2}$

$= \frac{4}{2}x + \frac{6}{2} \Rightarrow 2x+3$

$z = 2x+3.$

$dz = 2\,dx$

$$I = \int_{1}^{5}\frac{dz}{z} = \frac{b-a}{2}\int_{-1}^{1} f(z)\,dx = 2\int_{-1}^{1}\frac{dx}{2x+3}$$

$f(x) = \frac{2}{2x+3}.$

$f(-\sqrt{0.6}) = \frac{2}{2x-\sqrt{0.6}+3} = 1.3785$

$f(\sqrt{0.6}) = \frac{3}{2\times\sqrt{0.6}+3} = 0.4396$

$f(0) = \frac{2}{3}.$

$I = \frac{5}{9}\left[f(-\sqrt{0.6})+f(\sqrt{0.6})\right]+\frac{8}{9}f(0)$

$= \frac{5}{9}\left[1.3785+0.4396\right]+\frac{8}{9}\times\frac{2}{3}$

$= 1.6026$

Q) Evaluate $\int_{-2}^{2} e^{-\frac{x}{2}}\,dx$ by Gauss 2 point formula

$x = \frac{b-a}{2}z + \frac{b+a}{2}$

$= \frac{2-(-2)}{2}z + \frac{2+(-2)}{2} = \frac{4}{2}z = 2z$

$x = 2z.$

$dx = 2\,dz$

$$I = 2\int_{-1}^{1} e^{-\frac{2z}{2}}\,dz = 2\int_{-1}^{1} e^{-z}\,dz$$

$f(x) = 2e^{-z}.$

$f\left(\frac{-1}{\sqrt{3}}\right) = 2\times e^{\frac{-1}{\sqrt{3}}} = 1.1228$

$f\left(\frac{1}{\sqrt{3}}\right) = 2\times e^{1/\sqrt{3}} = 3.5626$

$2\int_{-1}^{1} e^{-z}\,dz = [1.1228+3.5626] = 4.6854$

(2) Evaluate $\int_0^{\pi/2} \sin t\,dt$ by Gaussian two point Formula.

$$t = \frac{b-a}{2}x + \frac{b+a}{2}$$

$$t = \frac{\pi}{4}x + \frac{\pi}{4} = \frac{\pi}{4}(x+1) \qquad t = \frac{\pi}{4}x + \frac{\pi}{4}.$$

$$dt = \frac{\pi}{4}dx.$$

$$I = \int_{-1}^{1} \sin\frac{\pi}{4}(x+1)\cdot\frac{\pi}{4}dx$$

$$= \frac{\pi}{4}\int_{-1}^{1} \sin\left[\frac{\pi x+\pi}{4}\right]dx.$$

$f(x) \sin\frac{\pi}{4}(x+1)$

$f\left(\frac{-1}{\sqrt{3}}\right) = \sin\frac{\pi}{4}\left(\frac{-1}{\sqrt{3}}+1\right)$

$= \sin(0.42265)$

$f\left(\frac{1}{\sqrt{3}}\right) = 1.57735$

$= \sin(1.57735)$

$$= \frac{\pi}{4}\left[\sin[0.42265] + \sin[1.57735]\right]$$

$$= \frac{\pi}{4}\left[0.7923 + 5.7936\times10^{-3} + 0.02162\right]$$

$f\left(\frac{-1}{\sqrt{3}}\right) = \sin[0.42265]$

$$= \frac{\pi}{4}\left[\sin + \sin(0.10566\pi)\right.$$

$$= \frac{\pi}{4}\left[\sin(0.10566\pi) + \sin(0.89434\pi)\right]$$

$$= \frac{\pi}{4}(0.99847)$$

$$= 0.99847$$

Euler's Algorithm

$$y_{n+1} = y_n + hf(x_n, y_n) \qquad n = 0,1,2\ldots$$

$$\frac{dy}{dx} = f(x,y) \qquad y(x_0) = y_0$$

Modified Euler's Formula.

$$y_{n+1} = y_n + h\left[f\left(x_n + \frac{1}{2}h, y_n + \frac{1}{2}hf(x_n,\right.\right.$$

Or.

RK method of 2nd order.

RK method or Runge-Kutta method of 4th order

$$k_1 = h(f(x_0, y_0);$$

$$k_2 = hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_1\right).$$

$$k_3 = hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_2\right).$$

$$k_4 = hf(x_0 + h, y_0 + k_3).$$

$$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$y_1 = y(x_0 + h) = y_0 + k$$

RK method of 2nd order (same as modified euler method).

$$k_1 = hf(x,y)$$

$$k_2 = hf\left(x + \frac{1}{2}h, y + \frac{1}{2}k_1\right)$$

$$\Delta y = k_2 \qquad h = \Delta x.$$

Q) Given $y' = -y$ and $y(0) = 1$, determine the value of y at $x = (0.02)(0.02)(0.04)$ by Euler method.

→ $y' = \frac{dy}{dx} = -y = f(x, y)$.

$y(0) = 1$

$y(x_0) = y_0$.

$x_0 = 0$

Take $h = 0.01$

$x = 0.2, 0.4$

$x = 0.1$

$x_0 = 0, y_0 = 1$

$x_1 = 0.01$

$x_2 = 0.02$

$x_3 = 0.03$

$x_4 = 0.04$

we have to find y_1, y_2, y_3, y_4.

if $x = 0.2, 0.4$.
choose $h = 0.2$.
$y = y_0, y_1, y_2$ error more
$h = 0.1$ error less
$x = 0, 0.1, 0.2, 0.3, 0.4$
$y = y_0, y_1, y_2, y_3, y_4$

$y_{n+1} = y_n + h f(x_n, y_n)$

$y_1 = y_0 + h f(0, 1)$

$= 1 + 0.01 \times (-1) = 1 - 0.01 = 0.99$.

$y_2 = y_1 + h f(x_1, y_1)$

$y_2 = 0.99 + 0.01 \times f(0.01, 0.99)$

$= 0.99 + 0.01 \times -0.99$

$= 0.9801$

$y_{n+1} = y_n + h f(x_n, y_n)$

$y_3 = y_2 + h f(x_2, y_2)$

$= 0.9801 + 0.01 \times f[0.02, 0.9801]$

$= 0.9801 + 0.01 \times -0.9801$

$= 0.9703$

$y_4 = y_3 + h f[x_3, y_3]$

$= 0.9703 + 0.01 f[0.03, 0.03]$.

$= 0.9703 + 0.01 \times -0.9703$

$= 0.9606$

Q) using Euler's method, solve numerically the equation $y' = x + y$, $y(0) = 1$ for $x = 0.0(0.2)(1.0)$

→ $h = 0.2$

$f(x, y) = x + y$.

$x_0 = 0$ $y_0 = 1$

$x_1 = 0.2$

$x_2 = 0.4$

$x_3 = 0.6$

$x_4 = 0.8$

$x_5 = 1.0$

Differentiation
Integration
3 methods for solving ordinary differential equation

$y_1 = y_0 + h f(x_0, y_0)$

$= 1 + 0.2 \times (0 + 1) = 1 + 0.2 = 1.2$

$y_2 = y_1 + h f(x_1, y_1)$

$= 1.2 + 0.2 \times [0.2 + 1.2] = 1.48$

$y_3 = y_2 + h f(x_2, y_2)$

$= 1.48 + 0.2 \times [0.4 + 1.48] = 1.856$

$y_4 = y_3 + h f[x_3, y_3]$

$= 1.856 + 0.2 [0.6 + 1.856]$

$= 2.3472$

$$y_5 = y_4 + h \times f[x_4, y_4]$$

$$= 2.3472 + 0.2 \times [0.8 + 2.3472]$$

$$= 2.97664$$

Modified Euler's Method Problems

Q) Compute y at $x = 0.25$ by modified Euler's method. given $y' = 2xy$ $y(0) = 1$

$y' = 2xy$

$f(x,y) = 2xy$. let $h = 0.25$

$x_0 = 0$ $y_0 = 1$

$x_1 = 0.25$ $y_1 = ?$

$$y_{n+1} = y_n + h\left[f\left(x_n + \frac{h}{2}, y_n + \frac{h}{2} f(x_n, y_n)\right)\right]$$

$$y_1 = y_0 + h\left[f\left(x_0 + \frac{h}{2}, y_0 + \frac{h}{2} f(x_0, y_0)\right)\right]$$

$$= 1 + 0.25\left[f\left(0 + \frac{0.25}{2}, 1 + \frac{0.25}{2} \times 0\right)\right]$$

$$= 1 + 0.25\, f\left[\left(\frac{0.25}{2}, 1\right)\right]$$

$$= 1 + 0.25 \times 2 \times 0.\frac{25}{2} \times 1$$

$$= 1.0625.$$

Q) Solve the equation $\frac{dy}{dx} = 1 - y$, given $y(0) = 0$ using modified euler's method at $x = 0.1$ and 0.2.

$f(x,y) = 1 - y$. let $h = 0.1$

$x_0 = 0$ $y_0 = 0$.

$x_1 = 0.1$ $y_1 = ?$

$x_2 = 0.2$ $y_2 = ?$

$$y_{n+1} = y_n + h\left[f\left(x + \frac{h}{2}, y + \frac{h}{2} f(x_n, y_n)\right)\right].$$

$$y_1 = y_0 + h\left[f\left(x_0 + \frac{h}{2}, y_0 + \frac{h}{2} f(x_0, y_0)\right)\right]$$

$$= 0 + 0.1\left[f\left(\frac{0.1}{2}, 0 + \frac{0.1}{2} \times 1\right)\right]$$

$$= 0 + 0.1 \times f\left(\frac{0.1}{2}, \frac{0.1}{2}\right) \quad (1 - 0.095)$$

$$= 0 + 0.1 \times 1 - \frac{0.1}{2}.$$

$$= 0.095.$$

$$y_2 = y_1 + h\left[f\left(x_1 + \frac{h}{2}, y_1 + \frac{h}{2} f(x_1, y_1)\right)\right]$$

$$= 0.095 + 0.1\left[f\left(0.1 + \frac{0.1}{2}, 0.095 + \frac{0.1}{2}(0.905)\right)\right]$$

$$= 0.095 + 0.1\left[f\left(0.15, 0.14025\right)\right]$$

$$= 0.095 + 0.1\left[1 - 0.14025\right]$$

$$= 0.180975$$

Q) Apply R-K method of 4th order to find $y(0.2)$ given that $y' = x+y$, $y(0)=1$ and $h=0.1$

$f(x,y) = x+y$.

$x_0 = 0 \quad y_0 = 1 \quad h = 0.1$

$x_1 = 0.1 \quad y_1 = ?$

$x_2 = 0.2 \quad y_2 = ?$

$k_1 = h[f(x_0, y_0)]$

$= 0.1 f(0,1)$

$= 0.1 \times 1 = 0.1$

$k_2 = hf\left[x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right]$

$= 0.1 f\left[0 + \frac{0.1}{2}, 1 + \frac{0.1}{2}\right]$

$= 0.1 f\left[\frac{0.1}{2}, \frac{2.1}{2}\right]$

$= 0.1 \left[\frac{0.1}{2} + \frac{2.1}{2}\right] = 0.1 \times \frac{2.2}{2} = 0.11$

$k_3 = hf\left[x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right]$

$= 0.1 f\left[0 + \frac{0.1}{2}, 1 + \frac{0.11}{2}\right]$

$= 0.1 \left[\frac{0.1}{2} + \frac{2.11}{2}\right] = 0.1 \times \frac{2021}{2} = 0.1105$

2.11
0.1
2.21

$k_4 = hf(x_0 + h, y_0 + k_3)$

$= 0.1 f[0 + 0.1, 1 + 0.1105]$

$= 0.12105$

$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$

$= \frac{1}{6}[0.1 + 2 \times 0.11 + 2 \times 0.1105 + 0.12105]$

$= 0.11034$

$y_1 = y_0 + k$

$= 1 + 0.11034 = 1.1103$

$y_2 = y_1 + k$.

$k_1 = h[f(x_1, y_1)]$

$= 0.1 f(0.1, 1.1103)$

$= 0.1 \times [0.1 + 1.1103] = 0.12103$

$k_2 = hf\left[x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2}\right]$.

$= 0.1 f\left[0.1 + \frac{0.1}{2}, 1.1103 + \frac{0.12103}{2}\right]$

$= 0.132 D8 = 0.1321$.

$k_3 = hf\left[x_1 + \frac{h}{2}, y_1 + \frac{k_2}{2}\right]$

$= 0.1 f\left[0.1 + \frac{0.1}{2}, 1.1103 + \frac{0.1321}{2}\right]$

$= 0.1326$

$k_4 = hf[x_1 + h, y_1 + k_3]$

$= 0.1[0.1 + 0.1, 1.1103 + 0.1326]$

$= 0.1443$

$$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$= \frac{1}{6}(0.1210 + 2\times.1321 + 2\times.1326 + 0.1443)$$

$$= 0.13245$$

$$y_2 = y_1 + k$$

$$= 1.1103 + 0.13245 = 1.24275 = 1.2428$$

* Using RK method of 4th order, find $y(0.7)$

If $y' = y - x^2$, $y(0.6) = 1.7379$

→ $f(x,y) = y - x^2$

$x_0 = 0.6$ $y_0 = 1.7379$. $h = 0.1$

$x_1 = 0.7$ $y_1 = ?$

$y_1 = y_0 + k.$

$k_1 = h f(x_0, y_0)$

$= 0.1 f(0.6, 1.7379)$

$= 0.1 [1.7379 - 0.6^2] =$

$= 0.13779 = 0.1378$

$k_2 = h f\left[x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right]$

$= 0.1 f\left[0.6 + \frac{0.1}{2}, 1.7379 + \frac{0.1378}{2}\right]$

$= 0.1 f[0.65, 1.8068]$

$= 0.1 [1.8068 - 0.65^2]$

$= 0.1384$

$k_3 = h f\left[x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right]$.

$= 0.1 f\left[0.6 + \frac{0.1}{2}, 1.7379 + \frac{0.1384}{2}\right]$

$= 0.1 f[0.65, 1.8071]$

$= 0.1 [1.8071 - 0.65^2]$

$= 0.1385$

$k_4 = h f[x_0 + h, y_0 + k_3]$

$= 0.1 f[0.6 + 0.1, 1.7379 + 0.1385]$

$= 0.1 f[0.7, 1.8764]$

$= 0.1 \times [1.8764 - 0.7^2] = 0.1386$

$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$

$= \frac{1}{6}(0.1378 + 2\times0.1384 + 2\times0.1385 + 0.1386)$

$= 0.13836 = 0.1384$

$y_1 = y_0 + k$

$= 1.7379 + 0.1384 = 1.8763$

Taylor series Method

→ To find numerical solutions of ODE, $\frac{dy}{dx} = f(x,y)$ with initial conditions, $y(x_0) = y_0$, by taylor series method.

→ $y_{n+1} = y_n + hy_n' + \frac{h^2}{2!}y_n'' + \frac{h^3}{3!}y_n''' + \cdots$

a) Solve $\frac{dy}{dx} = x+y$, $y(1)=0$ at $x=1.1$

$f(x,y) = x+y = y'$

$x_0 = 1$ $y_0 = 0$ $h = 0.1$

b. $x_1 = 1.1$ $y_1 = ?$

$y_{n+1} = y_n + hy_n' + \frac{h^2}{2!}y_n'' + \frac{h^3}{3!}y_n''' + \cdots$

$y_1 = y_0 + hy_0' + \frac{h^2}{2!}y_0'' + \frac{h^3}{3!}y_0''' + \cdots$

$= 0 + 0.1 \times 1 + \frac{0.1^2}{2!} \times 2 + \frac{0.1^3}{3!} \times 2 + \cdots$

$= 0 + 0.1 + \frac{0.1^2}{2!} \times 2 + \frac{0.1^3}{3!} \times 2 + \cdots$

$= 0 + 0.1 + 0.01 + 3.34 \times 10^{-4} + \cdots$

$= 0.1103$

$y' = x+y$

$y_0' = x_0 + y_0$
$= 1$

$y'' = 1+y'$

$y_0'' = 1+y_0'$
$= 1+1 = 2$

$y''' = y''$

$y_0''' = y_0'' = 2$

Q) Using Taylor's method, find $y(0.2)$ and $y(0.4)$ correct to 4 decimal places. Given $\frac{dy}{dx} = 1-2xy$ and $y(0)=0$

$h = 0.2$

$x_0 = 0$ $y_0 = 0$

$x_1 = 0.2$ $y_1 = 0.1947$

$x_2 = 0.4$

$f(x,y) = 1-2xy$

$y' = 1-2xy$ $y_0' = 1$

$y'' = 0-2[x.y'+y.1] = -2[xy'+y]$ $y_0'' = -2[0] = 0$

$y''' = -2[x.y''+y'+y']$
$= -2[xy''+2y']$
$y_0''' = -4$

$y'''' = -2[xy'''+y''+2y'']$
$= -2[0+0+0] = 0$

$y_{n+1} = y_n + hy_n' + \frac{h^2}{2!}y_n'' + \frac{h^3}{3!}y_n''' + \cdots$

$y_1 = y_0 + 0.2y_0' + \frac{0.2^2}{2!}y_0'' + \frac{0.2^3}{3!}y_0''' + \frac{0.2^4}{4!}y_0''''$

$= 0 + 0.2 \times 1 + 0 + \frac{0.2^3}{3!} \times -4 + 0$

$= 0.1947$

$y' = 1-2xy$

$y_1' = 1-2\times0.2\times0.1947$

$y_1' = 0.9221$

$y_1'' = -2[xy_1'+y]$
$= -2[0.2\times0.9221 + 0.1947]$
$= -0.7582$

$y_1''' = -2[xy_1''+2y_1']$
$= -2[0.2\times-0.7582 + 2\times0.9221]$
$= -3.3851$

$y_1'''' = -2[0.2\times -3.3851 + 2\times -0.7582]$
$= 5.9032$

$y_2 = y_1 + hy_1' + \frac{h^2}{2!}y_1'' + \frac{h^3}{3!}y_1''' + \frac{h^4}{4!}y_1''''$

$= 0.1947 \times 0.2 \times 0.9221 + \frac{0.2^2}{2!} \times -0.7582 + \frac{0.2^3}{3!} \times -3.3851 + \frac{0.2^4}{4!} \times 5.9032$

$= 0.3598$

Numerical Solution of Ordinary differential Equation by Finite difference method

Using Finite difference method, $y_i' = \frac{y_{i+1} - y_{i-1}}{2h}$

$$y_i'' = \frac{y_{i-1} + y_{i+1} - 2y_i}{h^2}$$

Substitute these values in given DE by putting $i = 1, 2, 3 \ldots (n-1)$ and boundary conditions we get $(n+1)$ equations. By solving these $(n+1)$ equations we can find $y_1, y_2, \ldots y_n$

Q) Using Finite difference method, solve for y from the given DE.

$$\frac{d^2y}{dx^2} + y + 1 = 0 \text{ where } x \in (0,1) \text{ and}$$

the Bcs, $y(0) = y(1) = 0$ by taking i) $h = \frac{1}{2}$ ii) $h = \frac{1}{4}$

i) $h = \frac{1}{2}$,

	x_0	x_1	x_2
x :	0	$\frac{1}{2}$	1
y :	0,	,	0
	y_0	y_1	y_2

$$\frac{d^2y}{dx^2} + y + 1 = 0$$

$$\frac{y_{i-1} + y_{i+1} - 2y_i}{h^2} + y_i + 1 = 0$$

$$\frac{y_{i-1} + y_{i+1} - 2y_i}{\left(\frac{1}{2}\right)^2} + y_i + 1 = 0.$$

$$4(y_{i-1} + y_{i+1} - 2y_i) + y_i + 1 = 0.$$

$$4y_{i-1} + 4y_{i+1} - 8y_i + y_i + 1 = 0.$$

$$4y_{i-1} + 4y_{i+1} - 7y_i + 1 = 0$$

put $i = 1$.

$$4y_0 + 4y_2 - 7y_1 + 1 = 0$$

$$4 \times 0 + 4 \times 0 - 7y_1 + 1 = 0.$$

$$7y_1 = 1$$

$$y_1 = 1/7$$

(ii) $h = \frac{1}{4}$.

$x_0 : 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$